

IB MATH AA SL

Lesson 35

Differentiating Polynomials

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Calculus · 60-minute lesson

Syllabus SL 5.3 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. Write $x \cdot x \cdot x$ as a power.
2. What is the gradient function called?
3. Differentiate is to find the ____?

Answers (click to reveal)

1. x^3
2. the derivative
3. gradient / rate of change

What we're learning today

LEARNING OBJECTIVES

- 1 Apply**
the power rule for differentiation.
- 2 Differentiate**
sums of power terms.
- 3 Differentiate**
terms with coefficients and constants.
- 4 Rewrite**
roots and fractions as powers first.

SUCCESS CRITERIA — I CAN...

- ✓
use $d/dx(x^n) = n x^{n-1}$
- ✓
differentiate term by term
- ✓
know the derivative of a constant is 0
- ✓
rewrite $\sqrt{x}$ and $1/x$ as powers before differentiating

The power rule

THE RULE

$$\frac{d}{dx} x^n = n x^{n-1}$$

Bring down

multiply by the power

Reduce

subtract 1 from the power

Constant

the derivative of a number is 0

TERM BY TERM

Differentiate each term separately.

Coefficient

stays multiplied (e.g. $3x^2 \rightarrow 6x$).

+5

$\rightarrow 0$.

Differentiate a polynomial

QUESTION

Differentiate $y = 3x^4 - 2x^2 + 7$.

SOLUTION

Power rule on each term:

$$\frac{d}{dx} (3x^4) = 12x^3$$

$$\frac{d}{dx} (-2x^2) = -4x$$

Constant $7 \rightarrow 0$, so:

$$\frac{dy}{dx} = 12x^3 - 4x$$

Your turn — differentiate

ANSWER IN YOUR BOOK

1. $y = x^5$. Find dy/dx .
2. $y = 4x^3$. Find dy/dx .
3. $y = x^2 - 3x + 2$. Find dy/dx .

Answers (click to reveal)

1. $5x^4$
2. $12x^2$
3. $2x - 3$

Roots & reciprocals

REWRITE FIRST

$$\sqrt{x} = x^{1/2}, \quad \frac{1}{x} = x^{-1}$$

Rewrite

change roots and fractions to powers

Then

apply the power rule as normal

EXAMPLES

$$\sqrt{x}$$

$$= x^{1/2} \rightarrow \frac{1}{2}x^{-1/2}$$

$$1/x$$

$$= x^{-1} \rightarrow -x^{-2}$$

Care

negative and fractional powers.

Differentiate a root

QUESTION

Differentiate $y = \sqrt{x}$.

SOLUTION

Rewrite the root as a power:

$$y = x^{1/2}$$

Apply the power rule:

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2}$$

Exam-style question

A curve is $y = 2x^3 - 5x + 4$.

- (a) Find dy/dx . [3]
- (b) Find the gradient at $x = 1$. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $dy/dx = 6x^2 - 5$ (A1)(A1)(A1)
- (b) at $x = 1$: $6(1)^2 - 5 = 1$ (M1)(A1)

Common mistakes



Constant

The derivative of a constant is 0, not the constant.



Power then $\times$

Multiply by the power AND reduce it by 1.



Rewrite roots

Change $\sqrt{x}$ to $x^{(1/2)}$ before differentiating.



Coefficients

$3x^2$ differentiates to $6x$ (keep the $\times 3$).

Mixed practice

QUESTIONS

1. $y = x^7$. dy/dx ?

2. $y = 5x^2 - 4x$. dy/dx ?

3. $y = x^3 + 6$. dy/dx ?

4. $y = 1/x$. dy/dx ?

Answers (click to reveal)

1. $7x^6$

2. $10x - 4$

3. $3x^2$

4. $-x^{-2} = -1/x^2$

Plenary & exit ticket

KEY FORMULAS

$$\frac{d}{dx} x^n = n x^{n-1}$$

$$\frac{d}{dx} (\text{constant}) = 0$$

EXIT TICKET — before you leave

1. $y = x^4$. dy/dx ?

2. $y = 3x^2 + 2$. dy/dx ?

3. Derivative of 9?

Answers (click to reveal)

1. $4x^3$

2. $6x$

3. 0

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Differentiating polynomials.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 36

Tangents, normals & rates of change.

$$y - y_1 = m(x - x_1)$$